

Mapping Dimensions of Occupational Spaces

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Significance Statement

Abstract

Understanding how individuals cognitively situate occupations relative to each other is key to establishing the cognitive basis of initial consideration stages in occupational mobility, formulating appropriate workforce development policies, and designing effective job recommendation tools. However, it remains unclear if individuals implicitly navigate the occupational space systematically and if so, by using occupational features like skills, tasks or knowledge. We develop a novel study to assess these representations and study the relative salience of these three dimensions in the occupational choices of 568 US participants. Our approach comprises two customized forced-choice comparison tasks, an occupational ranking task, and self-reported measures of occupational similarity and transition beliefs and preferences. We find that XYZ.

Introduction

Workers are increasingly looking to change occupations and find alternatives for many reasons. In the same vein, employers are shifting towards skills-based approaches that expand the recruitment pool and find employees based on skill-fit. Governments and international organizations are devising frameworks that encourage a skills-based view to help situate displaced workers and satiate labor market demand. Moving between occupations is comprised of many stages, each stage characterized by structural and individual constraints, a

crucial but understudied initial stage that many workers go through is the construction of a options set that they could potentially move to.

[Limitations of Current Approaches: policy tools and career guidance systems typically rely on explicit skill matching or occupational classifications, but these may not capture how people actually perceive and organize occupational space. The gap: we lack empirical understanding of the psychological dimensions individuals use to navigate this complex landscape.]

Existing approaches consider occupations as bundles of items along a single dimension like skills or tasks and uses these conceptualization to compute measures of occupational distance and link them to important socio-economic outcomes like wages or segregation.

current occupational mobility research in economics and sociology uses skills mismatch and task mismatch frameworks to understand wage outcomes and other socioeconomic variables of interest. These studies add valuable insights into labor market dynamics and the returns to different transition types. However, they focus on outcomes of completed transitions—analyzing patterns after workers have already selected into new occupations. The mobility process comprises many stages, and one crucial initial stage remains understudied: how workers construct their set of occupational alternatives in the first place. It would be misplaced to make inferences on workers' perceived available choices based on what they have already chosen.

Mobility patterns observed in aggregate data reflect a combination individual-level consideration and choice processes and structural factors on the supply and demand sides of the labor market. If workers vary in how they mentally organize occupational space—some by skills, others by knowledge or activities—then the same labor market structure might present different perceived boundaries to different individuals. Understanding these individual-level dimensional representations provides the micro-foundations for macro-level mobility patterns and can explain why some occupational boundaries remain permeable for certain workers while becoming rigid for others.

Existing work focuses on realized transitions and their outcomes—analyzing patterns after workers have already selected into new occupations. The mobility process comprises

many stages, and one crucial initial stage remains understudied: how workers construct their set of occupational alternatives in the first place, and which dimensions structure their perception of what alternatives exist.

The Consideration Set Problem

[Dive into the specific challenge: with thousands of possible occupations, how do workers even construct a feasible set of alternatives to consider? This is a fundamental decision-making bottleneck that shapes all downstream mobility. Explain why the initial consideration stage matters: workers can only pursue transitions they can imagine. The consideration set acts as a gateway—if an occupation isn’t in someone’s mental set of possibilities, no amount of objective skill-fit matters. This initial filtering shapes everything downstream: search behavior, training investments, and ultimately transition outcomes.]

Occupational Representations

To understand the consideration set evaluation, we need to understand how people mentally represent the occupational space. Workers may or may not compare occupations on all possible dimensions related to work simultaneously. In this paper, we posit that they likely use a reduced set of latent dimensions that organize their perception of what occupations are similar. These dimension-driven representations are the cognitive foundation that enables workers to generate and evaluate alternatives, critically shape which transitions workers contemplate.

[This Study - Present your approach and contribution: novel embedding-based methods to empirically map individual-level dimensional representations of occupational space, tested across diverse populations. Just 2-3 sentences explaining why these three dimensions are theoretically or practically meaningful. For example: "We focus on three core dimensions from the O*NET framework—Skills, Knowledge, and Activities—which represent related but distinct facets of occupational content that workers can directly observe and compare (e.g., 'what I do' vs. 'what I know' vs. 'what I'm capable of'). These dimensions span both the cognitive inputs and behavioral outputs of work, potentially capturing different bases for occupational comparison. "Preview that you find [key result about dimensional

structure/heterogeneity/convergence].]

Why SKA?

Economics: the increasing concern about the labour market implications of technological change urges us to look in more detail at structure of work content.

Sociology: Additionally, if professional and occupational boundaries are defined through work content, potentially leading to structural barriers to entry and transition, it is important to examine who different features of work content influence individuals evaluation of occupational alternatives.

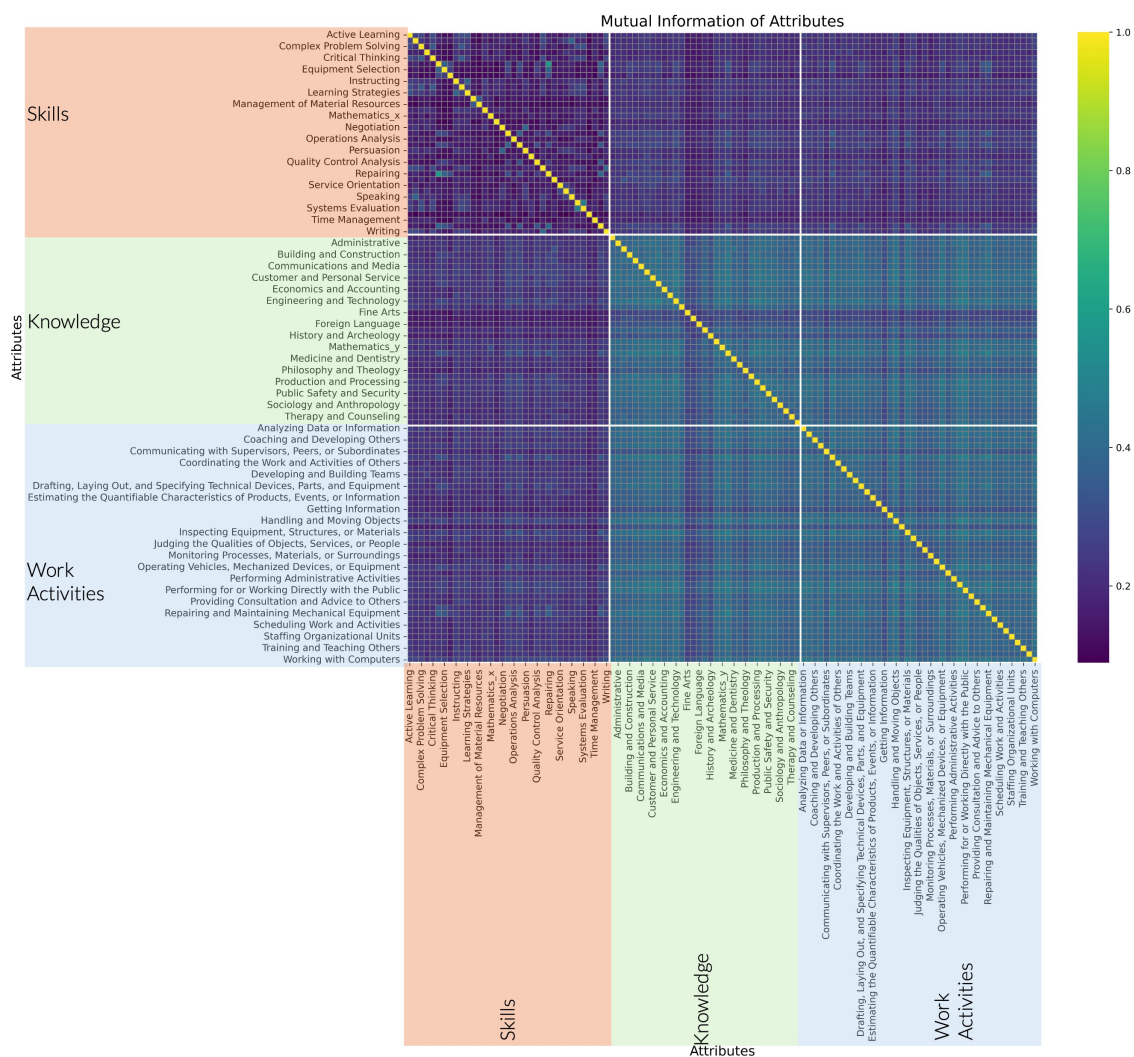


Figure 1: Mutual Information between Skills, Knowledge and Work Activities

Experimental Design

We developed novel ratings embedding-based tasks to assess the dimensions individuals to navigate occupational spaces. These include two forced choice occupational comparison tasks and an occupational ranking task. The tasks were uniquely tailored to a participant’s primary occupation and the tasks and questions were completed in a structured sequence within the survey that remained the same across participants. Initially, participants indicated their current or most recent role, which was matched to the Occupational Information Network (O*NET) database occupation titles using OpenAI’s embeddings. Participants selected the best match from a list of the five most semantically similar O*NET occupation. We use different tasks to obtain multiple within-subject measures of each respondent’s dimensional orientation when evaluating occupational alternatives, and also look at convergence across tasks. Both tasks reveal the attributes participants use to organize and carve up the occupational space.

Participants were recruited from two distinct sources. The first sample consisted of 130 participants recruited via email through NOVAworks, a regional Workforce Development organization in the United States. The second sample comprised 438 participants residing in the United States, recruited through Prolific, an online research participation platform. This dual recruitment strategy allowed us to capture perspectives from both individuals actively engaged with workforce development services and a broader online participant pool, providing complementary insights into how people from different contexts navigate and evaluate occupational alternatives. Refer to Table 1 for the distribution of participants across different categories. See Figure for a schematic of the study, figure for the oTree implementation and refer to the supporting information for technical details.

We argue that if an individual consistently selects the occupation that is more similar on a particular dimension, they see occupational intelligence in that dimension. While the forced-choice paradigm can be used isolate several different factors in a single task, to limit participant fatigue, we look for differences across only two dimensions at a time: skills-work activities, knowledge-work activities, and skill-knowledge.

Variable	Category	NOVAworks	Prolific
Participant Count	N	130	438
Gender	Female	82	209
	Male	47	227
	Other-Gender	1	2
Income	\$0-\$30,000	15	28
	\$31,000-\$60,000	10	102
	\$61,000-\$90,000	280	40
	\$91,000-\$120,000	135	26
	\$120,000+	164	29
Education	Less than high school diploma	2	1
	High school graduate or equivalent (eg: GED)	88	21
	Some college credit, no degree	162	34
	Bachelor's degree	481	80
	Trade/ Technical/ Vocational School	98	9
	Master's degree	182	39
	Doctorate degree	70	8
Ethnicity	Other	15	7
	White or Caucasian	726	116
	Black or African American	144	43
	Asian or Asian American	131	21
	Latino or Latina / Hispanic or Latinx	75	13
	American Indian or Alaska Native	11	4
	Native Hawaiian or other Pacific Islander	1	0
	Middle Eastern	2	1
Age	Other	8	1
	18-25	1	34
	26-35	10	148
	36-45	30	117
	46-60	61	113
	60+	28	26

Table 1: Demographic Breakdown of Participants

Analysis and Findings

First, we examined the distribution of participants' judgments across 30 trials, focusing on the differences in the number of choices oriented towards one attribute or another. Participants are categorized based on the total number of times they chose one of the two attribute neighbors in each of the three different studies. Note that the total number of attribute-1 neighbor and attribute-2 neighbor choices always adds up to 30. Specifically, participants are classified as having a strong orientation if they made at least 20 choices (i.e., two-thirds of the trials)

towards either attribute in any of the studies, categorizing them as "skills-strong," "activities-strong," "knowledge-strong," etc., depending on the attribute pair in question.

We conducted three separate studies, each utilizing a distinct occupational embedding space defined by a specific pair of attributes: skills and work activities, knowledge and skills, and knowledge and work activities. To evaluate participants' preferences systematically, we performed individual-level binomial tests. Each participant faced 30 binary decisions between two job types, with each decision treated as a Bernoulli trial in a sequence of 30 trials. The binomial test is used to determine whether the observed number of choices significantly deviates from what would be expected by chance, assuming a 50% probability for each attribute. Under the null hypothesis that there are no systematic cognitive orientations along any occupational attribute, choices would be expected to occur randomly, with no consistent bias toward either attribute. For 30 trials, a participant choosing an attribute-neighbor 20 or more times would exceed the expected random variation, indicating a significant orientation towards that attribute at the 5% significance level.

The results reveal that participants prioritize different occupational dimensions in all studies, as shown in figures 2, 3 and 4. In the study comparing skills and work activities, 27.6% of participants demonstrated a significant orientation towards skills, while 32.8% demonstrated a significant orientation towards work activities. In the study comparing skills and knowledge, 13.6% of participants showed a significant orientation towards skills, whereas 67.7% demonstrated a significant preference for knowledge. Finally, in the study comparing knowledge and work activities, 60% of participants were significantly oriented towards knowledge, while 13% preferred work activities.

Overall, more than 60% of participants in each of the three studies exhibited strong preferences for different occupational dimensions when constrained to specific occupational spaces, highlighting diverse cognitive orientations towards occupational attributes.

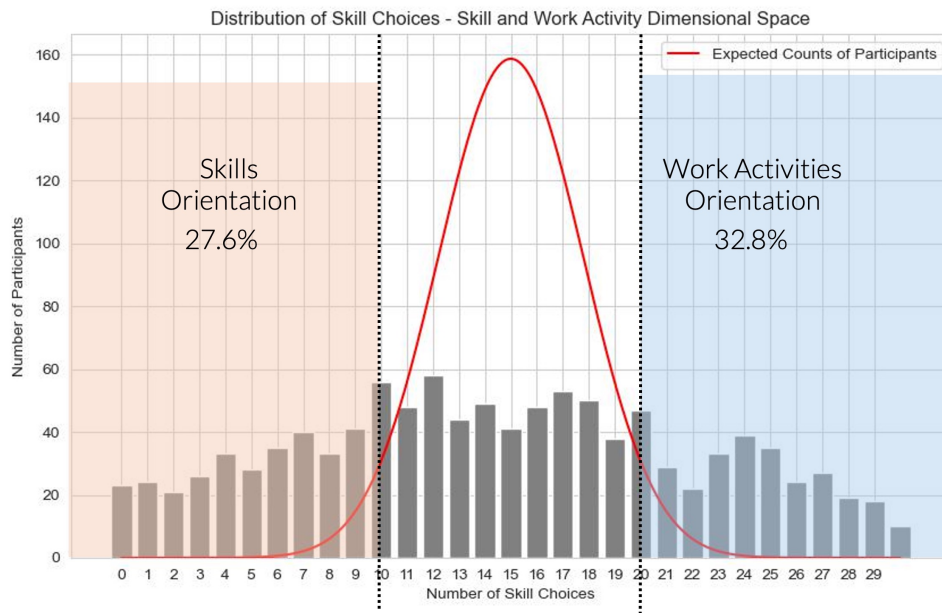


Figure 2: Skill and Work Activity Dimensional Space

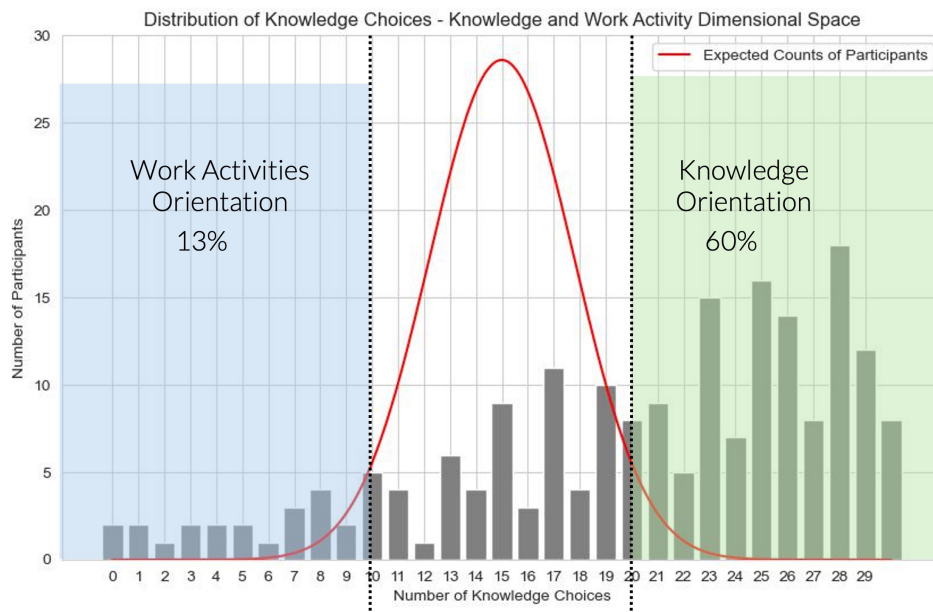


Figure 3: Knowledge and Work Activity Dimensional Space

Conclusions

When we consider different dimensions, the structure of the occupational space is more or less constrained depending on the dimension. The skills dimension carves occupational space more broadly implying that more occupations are skill-similar to any given occupation.

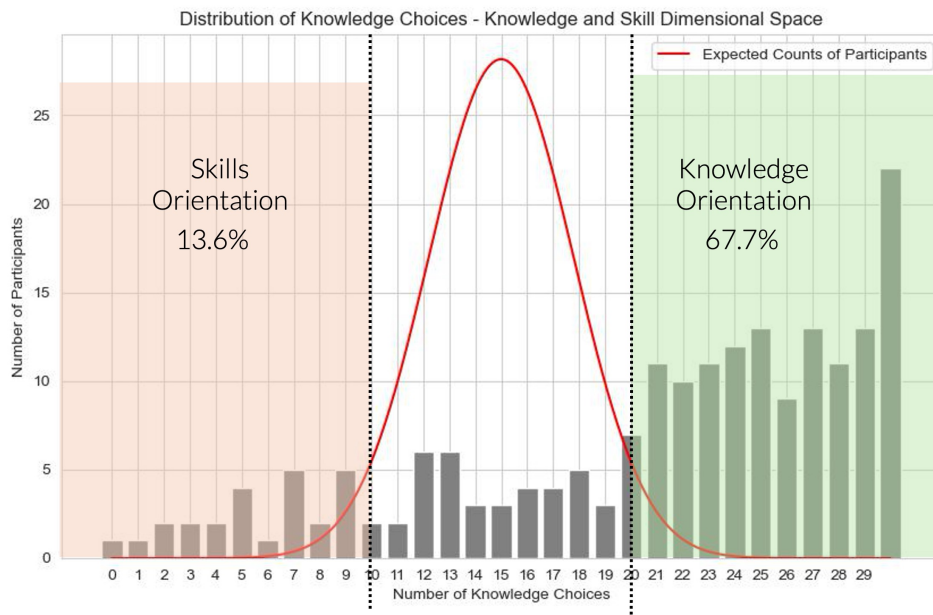


Figure 4: Knowledge and Skill Dimensional Space

Knowledge/activities dimensions are more constraining implying fewer occupations are similar along these dimensions. Therefore, workers who organize by knowledge/activities artificially limit their consideration sets. They miss out on a richer set of skill-similar alternatives that would yield better outcomes.

Existing models implicitly assume workers navigate occupational space using skill-based representations. However, if workers systematically organize occupations along alternative dimensions—such as knowledge or work activities—then their consideration sets may be systematically narrower, including occupations that appear well-matched along their salient dimension but are actually skill-distant. The skills dimension tends to identify a larger and more diverse set of similar occupations than knowledge or activity dimensions, as skills are more transferable across occupational boundaries. Workers who organize by knowledge or activities may therefore construct restricted consideration sets, overlooking skill-similar alternatives that would offer better economic returns.

Planned Next Steps

In future iterations of the study, we plan to expand the neighborhood scope to include both local and distant occupations. Instead of considering only occupations that are close to the focal job, participants will also be presented with pairs pertaining to an occupation that is distant from their focal occupation. This will allow the investigation of participants' ability to evaluate distant occupational alternatives and determine if different attributes are salient when considering occupations further removed from their experience. We also intend to introduce an additional occupational attribute, 'abilities,' distinct from skills, knowledge, and work activities, to further explore how individuals perceive occupational similarity based on a broader range of attributes. Additionally, we plan to extend the study to a within-subject design. This may involve two approaches: one in which participants evaluate all three attribute combinations (skills-work activities, skills-knowledge, and knowledge-work activities) for their focal occupation, and another in which participants evaluate both local and distant neighborhoods for a specific attribute combination.

Another line of inquiry involves prompting large language models (LLMs) to make these comparisons for all 873 occupations in the O*NET dataset across all attribute combinations. By leveraging LLMs, we can generate automated assessments of occupational similarity, providing a computational benchmark to compare against human participant responses. This approach will also help determine whether LLMs exhibit a consistent orientation toward specific occupational attributes, similar to or different from human participants.

References

Chen, D. L., Schonger, M., & Wickens, C. (2016). oTree-An open-source platform for laboratory, online, and field experiments. *Journal of Behavioral and Experimental Finance*, 9, 88–97. Publisher: Elsevier B.V.

Supporting Information

1. Occupational Data

1.1 ONET - what is it and why we use it

O*NET is particularly suitable for your study because it provides detailed, structured information across multiple dimensions (Skills, Knowledge, Activities), which is exactly what you need to test dimensional organization. Other systems are either less granular (ISCO, SOC) or proprietary/expensive (Lightcast).

1.2 Ratings

O*NET classifies skills, knowledge and work activities into 35, 33 and 41 categories respectively that occur across on all occupations in its database. Each skill is rated on a scale of 0-5 by O*NET occupational analysts. Similarly, occupational incumbents rate each knowledge domain and work activity on a scale of 0-5. All scales are averaged over multiple raters and standardized to a scale ranging from 0-100 to facilitate user comprehension. These ratings indicate the degree of importance of that skill, knowledge or work activity for a particular occupation. We retrieved the ratings for skills, knowledge and work activities ratings for each occupation from the O*NET database and converted these ratings into three separate vectors for each occupation which are used to construct embeddings that locate occupations in different spaces.

1.3 Occupational Similarity Measure

Each occupation, designated as the 'focal occupation' is compared with all others in the O*NET database, termed as 'target occupations'. For every occupation, we computed the cosine similarity between its skills vector and the skills vectors of all other occupations. This process was repeated for knowledge and work activities vectors. Target occupation pairs or neighbors are formed based on the computed closeness in an attribute combination. For example, if the combination is skills and work activities, option pairs for a focal occupation are constructed using these vector spaces in two stages. In the first stage, The occupations are embedded in skills and work activities vector spaces where each occupation is a point in these spaces. One space represents them with skill vectors where each vector element represents the human-rated importance of a specific skill for an occupation and the other space represents them similarly with human-rating vectors for work activities. Occupations in each of these spaces are closer or farther from each other based on their skill/work activity vectors. We used the cosine similarity measures of occupational closeness to construct a database of occupation pairs with their respective activity distance and skill distance measures. Occupations with similar skill requirements will be closer to the focal occupation in the skill space, and those with similar work activities will be closer in the activity space. Thus, for every occupation (focal occupation) in the O*NET database, a space can be constructed with all possible pairings with other occupations (target occupations) defined by two distances - skill and activity dissimilarity. This process is repeated for other attribute combinations - skills-knowledge and knowledge-work activities.

1.4 Forced Choice Task Option Criteria

In the second stage, for a particular focal occupation, I restrict the target occupations to a neighborhood in its space and consider pairs of target occupations within this neighborhood. For the example attribute combination of skills-work activities, two types of neighbors are defined based on skill and activity distances within this neighborhood. A skill neighbor is a target occupation that has a relatively low skill distance and relatively high activity distance, while an activity neighbor is a target occupation that has a relatively high skill distance and

relatively low activity distance. In other words, a 'skill neighbor' is an occupation that is close in skill space but distant in activity space to the focal occupation. Conversely, an 'activity neighbor' is close in the activity space but distant in the skill space. The neighborhood is defined by skill distance and activity distance thresholds (which vary by occupation) such that given a focal occupation's space of occupations, I find 15 neighbors of each type. Figure 5 illustrates graphically how neighbors are defined for the example of "Graphic Designers" as a focal occupation within its space of alternatives for the skills-work activities attribute combination.

1.5 Ranking Task Option Criteria

To generate the ranking set, we began by compiling three parent target lists corresponding to the key dimensions: skills, activities, and knowledge. These lists included all occupations from the ONET database, excluding the source (focal) job. Each parent list initially contained jobs that were the closest to the focal occupation based on cosine similarity of ratings of the respective dimensions' vectors from the ONET database. The initial step involved selecting the top 10 jobs for each dimension: skills, activities, and knowledge. Any job appearing in multiple lists was completely removed to maintain distinctiveness among the categories. After filtering out these duplicates, we selected the top three closest jobs from each dimension to form the final set, resulting in exactly nine unique jobs. In instances where the initial selection of the top 10 jobs per dimension did not yield nine unique jobs due to the removal of duplicates, we extended the consideration to the top 15 jobs for each dimension. By including a broader range of jobs, this step ensured that we had enough unique occupations to select three from each dimension. This method maintained a balance between providing clear distinctions among job dimensions and retaining representativeness. Ultimately, this careful construction of the final master list aimed to provide a well-defined and diverse set of occupations, allowing participants to rank the jobs meaningfully based on their prioritization of skills, activities, or knowledge. After constructing the 9-item set, the on-screen order is randomized for each participant to minimize order-based bias.

2. Survey Design

There are six major sections to the survey.

2.1 Occupational Title matching

Participants provided their current or most recent role as text input. We used openAI embeddings in real time to compare the textual cosine similarity of their occupational input and the entire collection of reported job titles for standard O*NET occupations. They were provided with a list of 5 semantically closest standard occupation titles to choose the most appropriate title that reflects their real-world current/past job. Participants had up to 5 opportunities to find a match. The title matching step was designed to ensure that the occupational comparisons presented to respondents in the forced-choice task are customized to their current/last-held job, preserving relevance to the participant's experience and avoiding the evaluation of random sets of occupations.

2.2 Two-Alternative Forced Choice Occupational Comparison Tasks

In the second section, the participant is asked to imagine a scenario where a colleague is considering changing occupations from the focal occupation and has approached the participant for advice. Based on the focal occupation, the participant is presented with 30 pairs of target occupations, one after the other and asked to choose the target occupation that is the closest match to the target occupation. These pairs of occupations are constructed based on the closeness of O*NET human rating vectors in the attribute-based embedding spaces as explained in the previous section. For the case of skill-work activity attribute combination, every triplet will consist of a focal occupation (that is an exact/close match to their current role), a skill neighbor, and an activity neighbor. Among the 30 pairs, target occupations that are skill neighbors do not overlap with activity neighbors. Target occupations repeat up to five times in the 30 pairs. The set of 30 pairs is fixed for every focal occupation, so any two participants who select a particular focal occupation will be faced with the same set of 30 pairs. The procedure is similar for the other attribute combinations. The schematic for the first two sections of the survey is shown in figure .

2.3 Occupational Ranking Task

2.4 Occupational Similarity and Transition Beliefs

2.5 Work Values and Occupational Transition Preferences

2.6 Demographics

In the third section of the survey, we collected additional information on participants' beliefs about factors that influence occupational similarity and occupational transition. Participants are asked to share information on demographic characteristics in the final section including age, gender, ethnicity, income and education level. The survey is built on the oTree platform (Chen et al., 2016) as conventional survey platforms were found to be inadequate for the specifics of the task design in title matching and comparison sections.

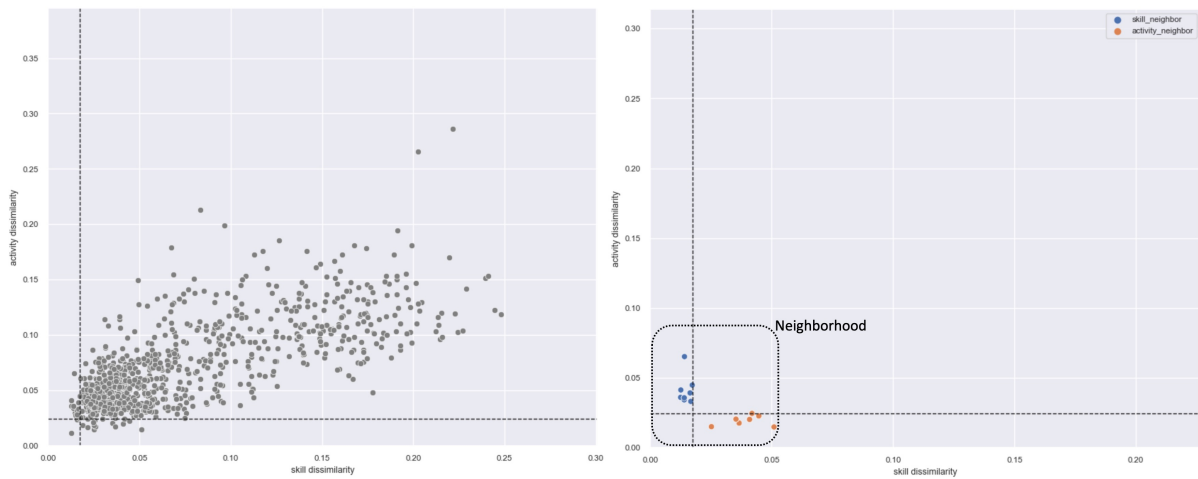


Figure 5: Pair Generation. x-axis: skill dissimilarity; y-axis: activity similarity

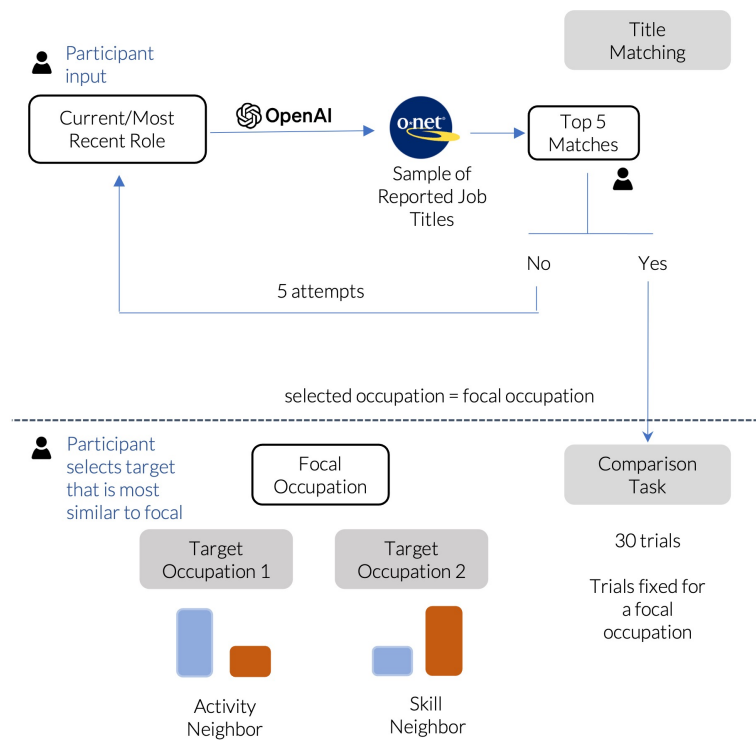


Figure 6: Survey Schematic

Title Matching Task	oTree Interface	Comparison Task
We would like to know about your current or recent role. We will try to match our input with occupations in our database. Using specific terms or the name you would use when first meeting someone will help us find the right match. In case you are independent or self-employed, provide a specific title that matches the kind of work you do. Searching our database might take a minute. Then, we will ask you to pick the closest match to your from a list of options. <i>If you don't see a good match, you can try submitting an alternate title for your role four more times.</i> What is your current or most recent role? Graphic Designer	Imagine you are mentoring a colleague who is considering changing occupations from the role of Graphic Designers and the colleague approaches you for advice. We will show you pairs of occupations and for each pair, you will be asked to give your opinion on which occupation is a closer match to the position currently held by the colleague. For every pair shown, use the left arrow key (←) to select the choice on the left and the right arrow key (→) to select the choice on the right.	
Please pick the closest match from the following list of options. <i>If you don't see a good match choose 'None of the above are close' and you can try submitting an alternate title for your role four more times.</i> Select the occupation that is most approximate to your response ☐ Commercial and Industrial Designers ☐ Art Directors ☐ Graphic Designers ☐ Special Effects Artists and Animators ☐ Web Developers ☐ None of the above are close	Imagine you are mentoring a colleague who is considering changing occupations from the role of Graphic Designers and the colleague approaches you for advice. Consider the following pair of occupations. Which job do you think is the closest match to the role of Graphic Designers? Lawyers Web Developers (←) left arrow (→) right arrow	

Figure 7: oTree Implementation